

Delong Zhou

Explore RNA Regulation in Disease Through RNA-Seq || Bridge Bioinformatics and
Experimental Biology

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Education

Ph.D. in Bioinformatics 2013-2022

University of Sherbrooke, Sherbrooke, QC

Engineering 2008-2013

École Centrale Paris, Île-de-France, France

Research Experience

Post doc in RNA Regulation 2025-Present

Lecuyer Lab of RNA Biology, Institut de Recherches Cliniques de Montréal, Montreal, QC

- Build and validate machine learning models to predict RNA localization based on RNA sequence
- Design webtool to visualize RNA Binding Protein motifs

Post doc in Psychology 2022-2024

Bagot Lab of Behavioural Neurogenomics McGill University, Montreal, QC

- Analyzed Single-Cell RNA-seq data of mouse brain samples
- Conducted behavior experiments in mouse depression model

Ph.D in Bioinformatics 2013-2022

Université de Sherbrooke; Sherbrooke, QC

- Analyzed RNA-seq datasets to predict RNA splicing and validated prediction using experimental approaches
- Developed a webtool to predict functional consequences of RNA alternative splicing
- Performed RNA-seq data processing and statistical analysis for collaborative projects, contributing to two peer-reviewed publications

Engineering 2011-2012

École Centrale Paris, Île-de-France, France

- Assisted in developing an autonomous potable water purification station in Cambodia

Institut Gustave-Roussy, Villejuif, France

- Studied cellular responses to nano-second electric shocks

Skills & Expertise

Language: English, French, Chinese (Mandarin, Wu)

Bioinformatics: Single Cell and Bulk RNA-seq analysis

Programming: Python, R, SQL, Bash, Linux, Git, HPC (SLURM)

Deep Learning: Tensorflow & Keras

Data Science: Pandas, numpy, duckdb, sqlite, matplotlib, seaborn; ggplot2

Cellular Biology: Cell culture, transfection, PCR, Western Blot, Immunoprecipitation

Laboratory animals: Mouse handling, intraperitoneal injection, behavioral tests

Academic Experiences

Mentoring, Supervision & Teaching

- Supervised student internships 2023-Present
- Taught undergraduate course Bioinformatics (2 credits, 25 students) 2021

Leadership & Academic Service

- Reviewed manuscripts for peer review journals 2019-Present
- Supported RNA Society Annual Meeting as volunteer 2026
- Served on Riboclub Student Seminar Jury & Executive Committee 2016-2020

Awards

Postdoctoral Fellowship Award (\$50 000) 2023-2024

McGill-Douglas & Max Planck Institute of Psychiatry.

Travel award (€500) 2024

European Behavioural Pharmacology Society Biennial Workshop. Banff, AB.

Travel award (\$500) 2019

Ribowest. Winnipeg, MB.

Trophy of Innovation and Sustainable Developments (€6000). 2012

Paris, France.

Community Involvement

- Participated in outreach Concerts 2026-Present
- Volunteered at Kahéhtaïen Jardin Lumb 2026-Present